

A multi-team benchmark of AI approaches for predicting behavioral-experiment effects

2026-06-09

This is an amendment to our preregistration “*Predicting the effects of behavioral interventions on humans with LLMs*”. Writing the initial pre-registration, we conceived of the project as a single-team simulation project. With this preregistration amendment, we intend to turn the project into a **multi-team prediction tournament**: we will invite independent research teams to each predict the same human results using whatever AI-based simulation approach they prefer. The motivation is to catalogue the diversity of current AI approaches to forecasting behavioral experiments and to test, on genuinely novel data, how well they perform. The initial preregistration remains valid. It describes in detail how intend to measure the (dis)similarity between the human sample and LLM generated samples of “behavioral clones, and we will stick to all analytic procedures layd out there. We will report the simulations that we, the core team, have provided as part of the initial preregistration as one team’s approach.

At the point of writing this amandement to the pre-registration, human data is partially collected (~10,000 of the ~22,000 responses). Crucially, none of the invited teams will be granted access to the human data. Before any human data are released, all teams will need to preregister their simulation approach, following a standardized reporting form. As part of this process, the teams will need to make their simulated data sets public.

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```
# packages
library(tidyverse)
library(sandwich)
library(lmtest)
library(marginaleffects)
library(broom)
library(MetBrewer)
library(patchwork)
library(gt)
library(here)

# custom functions (reused unchanged from the original preregistration)
source(here("R/functions/statistics.R"))
source(here("R/functions/plots.R"))
source(here("R/functions/tables.R"))
source(here("R/functions/reporting.R"))

# simulated (random) placeholder data
human_data      <- readRDS(here("data/simulation_preregistration/human_data_preregistration.rds"))
llm_data_teams <- readRDS(here("data/simulation_preregistration/llm_data_preregistration_teams.rds"))

# Same preregistered random split of the human sample as in the original
# preregistration. Human 1 is the reference half all submissions are scored
# against; Human 2 is the human-human ceiling.
set.seed(42)
split_ids      <- sample(unique(human_data$id), size = floor(length(unique(human_data$id)) / 2))
human_data_1 <- human_data |> filter(id %in% split_ids)
human_data_2 <- human_data |> filter(!id %in% split_ids)
```

! Important

This is an amendment. It builds on, and does not replace, the original preregistration ([Preregistration](#)). All analytical procedures are kept in place. This document adds (i) the tournament design, (ii) a standardized reporting framework for describing each team’s approach, and (iii) a new cross-team comparison.

All datasets used in this amendment are simulated random noise with the correct structure.

Motivation

The original preregistration evaluates a single approach to simulating human survey responses with LLMs—our own behavioral clones, built from demographically representative U.S. profiles and a particular model and prompt. However, a single team’s result only represents one point in a large design space of simulating behavioral clones. A more systematic way of assessing how well behavioral clones can mimic human participants in survey experiments is to pool expertise and report different approaches. Since research on behavioral clone samples is very recent, there is additional value in outlining different approaches that researchers in the field take to obtain human-like samples.

We will therefore launch a call for a **multi-team prediction tournament**. Predicting a large survey experiment ($N = \sim 22,000$) designed to increase trust in climate scientists provides a unique benchmarking opportunity for such a tournament. The experiment tests the effects of 16 messages¹, and includes both self-reported attitudes (e.g., support for funding of climate science) and behavioral measures (e.g., donations to a scientific organization) as outcomes. Importantly, none of the interventions has ever been tested in that form. This matters, because the perhaps most discussed use-case of behavioral clones is simulating novel research paradigms that would otherwise be very costly to explore with human participants. The experiment is, at the point of writing, only partially conducted ($\sim 10,000$ of the $\sim 22,000$ responses), and none of the data will be available to the teams participating in the tournament. This overcomes the issue of researcher degrees of freedom when assessing the capacities of behavioral clones (Cummins 2025).

The first goal of the tournament is not to crown a single champion team—per the original pre-registration, we will evaluate the approaches along different metrics, and it seems likely that some approaches will do better on some metrics, but worse on others. Instead, we would like to provide a **collective benchmark** of how close behavioral clones get to mimicking human respondents across the different metrics, so there will likely be multiple “winners”. Second, and perhaps more importantly, we aim to provide a **catalogue**: a structured, comparable

¹The original experiment contains 20 interventions, but we will only use 16, as the other four are interactive interventions that would make the simulation process more complicated.

description of how research teams are currently approaching AI prediction of behavioral experiments. Ideally, comparing the different approaches and their results will allow us to speculate on how some design choices (e.g., open- vs. closed-weight models, persona source, one-shot vs. full-survey presentation) may affect the accuracy of the behavioral clones in mimicking humans.

Tournament design

The central methodological guarantee of this tournament is that **no participating team can observe the human target data**. The human outcome data are collected and held exclusively by the core team. At the time of registering this amendment, only about 10,000 of the planned ~22,000 human responses have been collected, and none of the outcome data have been shared, published, or otherwise made available.

All approaches must (i) be **AI-based**, i.e. not involve humans in the loop and (ii) produce **individual-level behavioral clones**: each submission is a dataset of synthetic respondents who each complete the experiment, conforming to the analysis schema (see *Submission requirements*). Experiment-level or aggregate-only predictions—a single effect estimate per condition, without underlying synthetic respondents—are out of scope, as are purely statistical models without an AI component.

Teams are otherwise **unrestricted**: they may use any AI models, any prompting or persona strategy, and any external training data or prior studies they wish.

We will send out invitation to researchers with a track record of working on behavioral clones. The procedure for each team is:

1. **Fill out a standardized reporting framework.**
2. **Generate behavioral clones data.**
3. **Preregister all**

Preregistration

Teams are required preregister (i) the standardized reporting framework which describes their simulation approach, and (ii) the behavioral clones data that they generated.

If possible, we encourage teams to also preregister their simulation code and other relevant materials.

Because it offers a convenient option to create preregistrations from [generating a release on a linked github repository](#), we will ask teams to register via [Zenodo](#).

Reporting framework

Every team describes its approach using the **GUIDE-LLM** reporting checklist (Feuerriegel et al. 2026), a recently published standard for reporting LLM use in the behavioural and social sciences. Teams complete the full checklist as part of their preregistration; below we reproduce the items relevant to generating behavioural clones, adapt their wording to this tournament, and add a small number of **tournament-specific items** (marked) that are essential for comparability here but are not part of GUIDE-LLM. Items that concern human participants, participant-facing interventions, or sensitive-data privacy are generally *not applicable*, because every submission is a fully synthetic dataset generated without human subjects; teams may mark these “N/A”. Sharing runnable code and materials on top of this description is strongly encouraged (see *Preregistration*).

Approach identity and output (*tournament-specific*)

- Team name and members.
- One-paragraph plain-language summary of the approach.
- Approach family within individual-level simulation (e.g., one LLM call per respondent vs. an agent that walks through the full survey; single model vs. multi-agent).
- Confirmation that the submission is an individual-level behavioural-clone dataset conforming to the analysis schema (see *Submission requirements*); number of synthetic respondents and how they map to conditions; coverage of outcomes and conditions (all, or a declared subset).

A. Scope of LLM use

- **A.1 — Purpose.** The stage(s) of the workflow where LLMs are used (here, at minimum, generating synthetic survey responses; note any additional use, e.g., constructing personas or post-processing outputs).
- **A.2 — Degree of automation.** Confirm the pipeline is fully automated with no human in the loop (a tournament requirement); describe any exception and who provided oversight.

B. Model / system details

- **B.1 — Model name(s).** Exact model name(s) including provider, model size, version/ID, date of access, and a source link where applicable (e.g., “*GPT-4o-mini-2024-07-18 (OpenAI)*”, “*Llama-3.1-8B (Meta, via HuggingFace)*”). If several models were tested, name them and state which produced the submission and why.
- **B.2 — Access and context mode.** How the model was accessed (API / web interface / local) and whether it was used in chat mode (ongoing conversation) or stateless mode (separate calls); exact API name and version.

- **B.3 — Configuration.** Settings affecting output: temperature, top-k / top-p, max tokens, frequency / presence penalties, stop sequences, number of completions or runs per respondent, quantization (e.g., FP16 / INT8 / INT4), and any reasoning-effort settings.
- **B.4 — Customization.** Any adaptation beyond standard inference: base model / fine-tuning (e.g., LoRA) / retrieval-augmentation (RAG) / automated prompt optimization / tool or function calling / web search / agentic workflows / post-training (RLHF, DPO) / other. Describe each one used.
- **B.5 — Persistent memory.** Whether sessions carried persistent memory across interactions (yes / no / N/A).

C. Prompts

- **C.1 — Exact prompts.** Verbatim prompt text(s), including any in-context examples, or a link to them in the preregistered repository. Document whether prompts were iteratively refined, and whether refinements were pre-specified or made in response to interim outputs.
- **C.2 — System-wide instructions.** Any system-level instructions guiding the model’s general behaviour.

Persona / profile construction *(tournament-specific)*

- Source of the demographic profiles: GSS / ANES / Census / other survey / fully synthetic / none.
- Variables encoded in each persona, and how they are injected into the prompt.
- Number of personas and their assignment to conditions (between- vs. within-subjects).
- Any weighting or matching to a target population; whether personas are reused across conditions.

Stimulus and survey presentation *(tournament-specific)*

- How the interventions / stimuli are presented to the model (verbatim vs. paraphrased or summarized).
- How survey items are presented (one item at a time vs. the whole survey; item and response-option ordering / randomization; how the response scales are shown).

E. Validation & interpretation

- **E.1 — Human validation.** Whether and how any human review of model outputs was performed (often N/A for fully automated generation).
- **E.2 — Post-processing.** How free-text outputs were parsed into the response scales; handling of refusals, malformed, missing, or out-of-range responses; how multiple generations were consolidated into the submitted values.

F. Reproducibility

- **F.1 — Materials shared.** Whether code, notebooks, prompts, and logs are shared (link / DOI), with API keys and any sensitive material removed; determinism and seeds.

G. Competing interests

- **G.1 — Relationships.** Funding, in-kind compute or model access, or professional / financial relationships with entities that have an interest in LLMs.

Data inputs and the blinding attestation (*tournament-specific*)

- A self-declared list of any external data that informed the approach (training, fine-tuning, retrieval, in-context examples, calibration anchors) and any prior experiments or public datasets used.
- Following GUIDE-LLM’s training-data-leakage item, a brief novelty note: the interventions are previously untested in this form and the human outcome data are unpublished and inaccessible, so leakage of the prediction target is structurally precluded.
- **A mandatory attestation that the team had no access to this study’s human outcome data.**

Recommended items (*GUIDE-LLM optional*)

- **Model-choice justification** along performance, transparency (open- vs. closed-weight; known training data / objective), reproducibility, ethics, and cost.
- **Prompt-design rationale** (structured format, established guidelines, prior studies, few-shot selection, standardization across models).
- **Comparison against other models / methods**, if any, and the benchmarking procedure used.
- **Risk of bias** or systematic differences in model behaviour across groups that could affect conclusions.
- **Computational resources:** API-call counts, total tokens, financial cost, and compute time.

Submission requirements

Every submission is an **individual-level behavioral-clone dataset**: a respondent-per-row dataset conforming to the cleaned analysis schema (the structure of `cleaned_llm.rds`—one row per synthetic respondent, with `condition` and all preregistered outcome and moderator columns). Because the submission has the same shape as our own clone data, it flows through the original analysis pipeline unchanged and is eligible for **all** analyses: average treatment effects (Section 1), subgroup heterogeneity (Section 2), and demographic baseline calibration (Section 3). Standard errors, inferential agreement, and equivalence tests are computed from the fitted models, so no separate uncertainty input is required from teams.

Experiment-level or aggregate predictions (a single effect estimate per condition, without underlying synthetic respondents) are not accepted (see *Tournament design*). Requiring one common individual-level schema is precisely what makes submissions directly comparable and lets the entire original analysis pipeline apply without modification.

The model functions in `R/functions/statistics.R` are agnostic to the specific condition labels and to the data source, so the same code path applies to human data, our own clones, and any team’s submission.

Analysis plan

Per-team analysis (reused)

Each submission is analyzed exactly as the original preregistration specifies, against the **Human 1** reference half. We do not restate that plan here; see the [original preregistration](#) for the full specification of:

- **Section 1 — Average Treatment Effects:** ATE recovery across all outcomes (`run_main_treatment_model()` / `run_main_treatment_model_binary()`, compared with `compare_estimates()` and `run_calibration()`), and control-condition response distributions (`compare_distributions()`).
- **Section 2 — Subgroup Effects:** condition $\times$ moderator heterogeneity across the six moderators (`run_moderator_model()`), and within-subgroup distributions.
- **Section 3 — Demographic Baseline Calibration:** group-mean calibration (`compare_demographic_baselines()`) and demographic predictability / stereotyping (`compare_demographic_predictability()`).

The **Human 1 vs. Human 2** split provides the human–human ceiling against which all teams are read, exactly as in the original document.

Cross-team comparison (new)

This is the new analytical contribution of the amendment. We assemble a **leaderboard**: each team is scored with the same metric ladder used throughout the original preregistration, and teams are ranked so that approaches can be compared directly and against the human–human ceiling.

For the headline leaderboard we pool average-treatment-effect estimate pairs across all continuous outcomes and report the comparison-metric ladder in increasing order of strictness—directional agreement, Spearman ρ , Pearson r , inferential agreement, and TOST equivalence—plus calibration slope and R^2 on the primary outcome. Because every submission is individual-level, the distribution (Sections 1b, 2b) and demographic-calibration (Section 3) metrics are likewise available for every team and added as further leaderboard columns. We then relate

these performance metrics to approach characteristics coded from the reporting form (e.g., open- vs. closed-weight model, fine-tuned vs. not, persona source, one-shot vs. full-survey presentation).

i Note

Exploratory by design. With a modest number of teams, the relationship between approach characteristics and predictive performance is reported **descriptively** (rankings, group comparisons, visual patterns). We do not preregister inferential tests of feature effects; the cross-team feature analysis is hypothesis-generating, intended to surface which design choices appear to matter for future, larger benchmarks.

The remainder of this section demonstrates the leaderboard code on simulated multi-team placeholder data: five mock “teams”, each a full LLM-style simulation drawn with a different seed and stacked with a `team` label. The code will be applied to real submissions without modification.

```
outcomes_primary <- "trust_multidimensional"
outcomes_secondary <- c("trust_post", "distrust_post",
                        "funding_perceptions", "policy_role_mean", "inst_trust_mean")
outcomes_tertiary <- c("belief_post", "concern_mean", "policy_general",
                      "policy_specific_mean", "behavior_mean")
outcomes_continuous <- c(outcomes_primary, outcomes_secondary, outcomes_tertiary)

outcome_label_map <- c(
  trust_multidimensional = "Multidimensional trust (primary)",
  trust_post             = "Overall trust",
  distrust_post          = "Distrust",
  funding_perceptions    = "Funding perceptions",
  policy_role_mean       = "Policy role",
  inst_trust_mean        = "Institutional trust",
  belief_post            = "Climate belief",
  concern_mean           = "Climate concern",
  policy_general         = "General climate policy",
  policy_specific_mean   = "Specific climate policy",
  behavior_mean          = "Pro-climate behavior"
)

# Extract ATE pairs (estimate, SE, adjusted p) - same helper as the original
# preregistration's rq1-all chunk.
prep_ate <- function(df) {
  df |>
    select(condition, estimate) |>
```

```

mutate(
  se      = if ("std.error" %in% names(df)) df$std.error      else NA_real_,
  p_adj   = if ("p.value_adjusted" %in% names(df)) df$p.value_adjusted else NA_real_
)
}

# Reference (Human 1) ATE models, fit once and reused for every comparison.
human_models <- map(outcomes_continuous,
  ~ run_main_treatment_model(human_data_1, outcome = .x)) |>
  set_names(outcomes_continuous)

# Pool ATE estimate pairs across continuous outcomes for one prediction dataset
# against the Human 1 reference models.
build_pairs <- function(pred_data) {
  map_dfr(outcomes_continuous, function(out) {
    inner_join(
      prep_ate(human_models[[out]]) |>
        rename(estimate_h = estimate, se_h = se, p_adj_h = p_adj),
      prep_ate(run_main_treatment_model(pred_data, outcome = out)) |>
        rename(estimate_l = estimate, se_l = se, p_adj_l = p_adj),
      by = "condition"
    ) |> mutate(outcome = out)
  })
}

# Six-metric ladder over pooled pairs - mirrors compare_estimates() /
# the pooled_row logic in the original preregistration.
pooled_metrics <- function(pairs) {
  pairs |>
    mutate(
      delta      = 0.5 * abs(estimate_h),
      diff       = estimate_h - estimate_l,
      se_diff    = sqrt(se_h^2 + se_l^2),
      p_tost     = pmax(pnorm((diff + delta) / se_diff, lower.tail = FALSE),
                        pnorm((diff - delta) / se_diff)),
      equivalent = p_tost < 0.05,
      same_sign  = sign(estimate_h) == sign(estimate_l),
      infer_h    = case_when(
        !is.na(p_adj_h) & p_adj_h < .05 & estimate_h > 0 ~ "pos",
        !is.na(p_adj_h) & p_adj_h < .05 & estimate_h < 0 ~ "neg",
        TRUE ~ "ns"),
      infer_l    = case_when(

```

```

      !is.na(p_adj_l) & p_adj_l < .05 & estimate_l > 0 ~ "pos",
      !is.na(p_adj_l) & p_adj_l < .05 & estimate_l < 0 ~ "neg",
      TRUE ~ "ns")
    ) |>
    summarise(
      directional_pct = mean(same_sign, na.rm = TRUE) * 100,
      spearman_rho    = cor(estimate_h, estimate_l, method = "spearman",
                           use = "pairwise.complete.obs"),
      pearson_r       = cor(estimate_h, estimate_l, use = "pairwise.complete.obs"),
      inferential_pct = mean(infer_h == infer_l, na.rm = TRUE) * 100,
      tost_pct        = mean(equivalent, na.rm = TRUE) * 100
    )
  }
}

team_levels <- levels(llm_data_teams$team)

# One leaderboard row per team
team_rows <- map_dfr(team_levels, function(tm) {
  pred <- llm_data_teams |> filter(team == tm)
  pairs <- build_pairs(pred)
  metrics <- pooled_metrics(pairs)
  calib <- run_calibration(human_models[[outcomes_primary]],
                         run_main_treatment_model(pred, outcome = outcomes_primary))

  metrics |>
    mutate(submission = tm,
           beta = calib$beta, calib_r2 = calib$r_squared)
})

# Human-human ceiling row (Human 1 vs. Human 2)
ceiling_pairs <- build_pairs(human_data_2)
ceiling_calib <- run_calibration(human_models[[outcomes_primary]],
                               run_main_treatment_model(human_data_2,
                                                         outcome = outcomes_primary))

ceiling_row <- pooled_metrics(ceiling_pairs) |>
  mutate(submission = "Human-human ceiling",
         beta = ceiling_calib$beta, calib_r2 = ceiling_calib$r_squared)

leaderboard <- bind_rows(ceiling_row, team_rows) |>
  arrange(desc(submission == "Human-human ceiling"), desc(pearson_r))

```

```

leaderboard |>
  select(submission, directional_pct, spearman_rho, pearson_r,
         inferential_pct, tost_pct, beta, calib_r2) |>
  gt() |>
  fmt_number(columns = c(spearman_rho, pearson_r, beta, calib_r2), decimals = 2) |>
  fmt_number(columns = c(directional_pct, inferential_pct, tost_pct), decimals = 0) |>
  cols_label(
    submission      = "Submission",
    directional_pct = "Directional %",
    spearman_rho    = "Spearman ",
    pearson_r       = "Pearson r",
    inferential_pct = "Inferential %",
    tost_pct       = "TOST %",
    beta           = "Calib. ",
    calib_r2       = "Calib. R2"
  ) |>
  tab_style(
    style      = cell_fill(color = "#eef3f7"),
    locations = cells_body(rows = submission == "Human-human ceiling")
  )

```

Table 1: Cross-team leaderboard (placeholder data). Pooled ATE-recovery metrics across all continuous outcomes, plus calibration slope and R² on the primary outcome, scored against the Human 1 reference half. The Human–human ceiling (Human 1 vs. Human 2) is the upper bound for any approach. Metrics run least-to-most strict, left to right.

Submission	Directional %	Spearman	Pearson r	Inferential %	TOST %	Calib.	
Human–human ceiling	48	-0.10	-0.06	98	0	-0.16	
team_5	58	0.27	0.33	99	0	-0.09	
team_1	52	0.06	0.11	99	0	-0.28	
team_2	42	-0.15	-0.10	99	0	-0.42	
team_3	50	-0.07	-0.11	99	0	-0.34	
team_4	50	-0.13	-0.18	99	0	0.03	

```

ceiling_r <- leaderboard |> filter(submission == "Human-human ceiling") |> pull(pearson_r)

leaderboard |>
  filter(submission != "Human-human ceiling") |>
  mutate(submission = fct_reorder(submission, pearson_r)) |>
  ggplot(aes(x = pearson_r, y = submission)) +

```

```
geom_vline(xintercept = ceiling_r, linetype = "dashed", colour = "grey40") +
geom_segment(aes(x = 0, xend = pearson_r, yend = submission), colour = "grey80") +
geom_point(size = 3, colour = met.brewer("Egypt")[2]) +
annotate("text", x = ceiling_r, y = 0.6, label = "Human-human ceiling",
         hjust = 1.05, size = 3, colour = "grey40") +
labs(x = "Pearson r (pooled ATEs vs. Human 1)", y = NULL) +
theme_minimal()
```

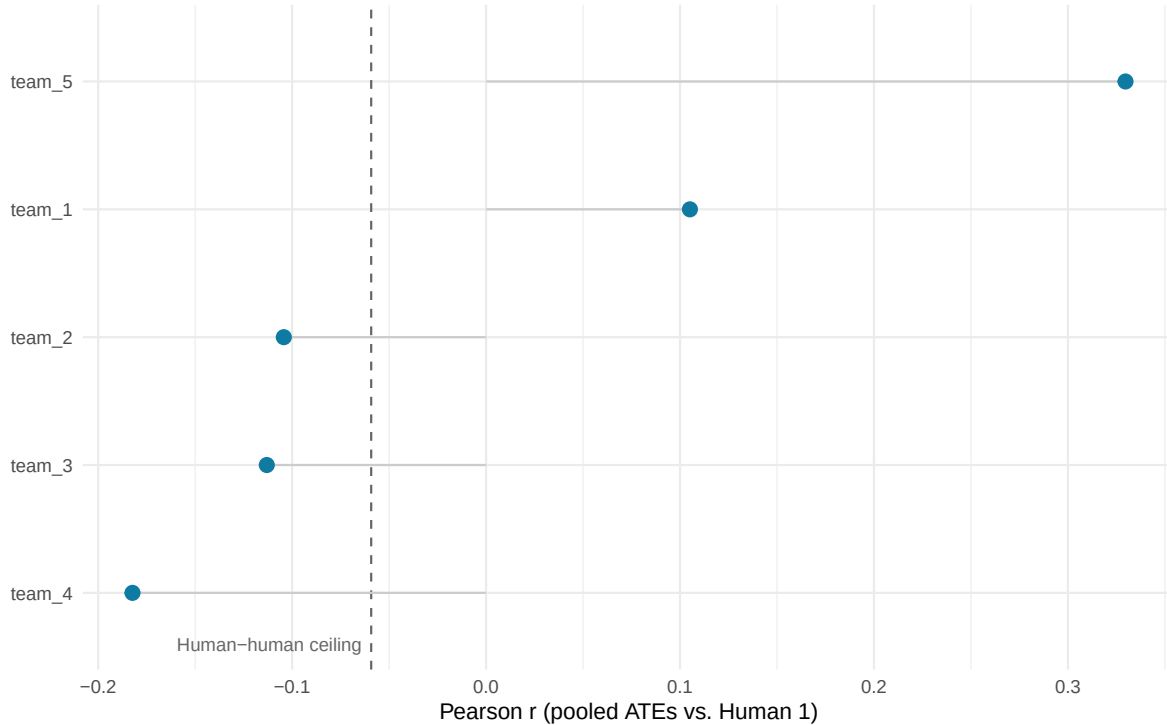


Figure 1: Approaches ranked by ATE-recovery (Pearson r of pooled ATEs against Human 1), with the human–human ceiling marked. On placeholder data all submissions are random noise, so no approach is expected to separate from the others.

When real submissions arrive, this leaderboard is recomputed by replacing the placeholder team dataset with the stacked set of submitted clone datasets, and by adding the distribution and demographic-calibration metrics from Sections 1b, 2b, and 3 as additional leaderboard columns (available for every submission, since all are individual-level). The approach-characteristics table coded from the reporting form is then joined to the leaderboard for the exploratory analysis of which design choices accompany better predictions.

Links and registration

- **Original preregistration (this project):** [Preregistration](#).
- **Human study preregistration:** <https://doi.org/10.5281/zenodo.20160212>.
- **Core-team LLM simulation data:** <https://doi.org/10.5281/zenodo.20167059>.
- **Simulation code (time-stamped release):** <https://github.com/yarakyrychenko/llm-participants/releases/tag/v1.0.0>.

Cummins, Jamie. 2025. “The Threat of Analytic Flexibility in Using Large Language Models to Simulate Human Data.” arXiv. <https://doi.org/10.48550/ARXIV.2509.13397>.

Feuerriegel, Stefan, Christopher Barrie, Molly J. Crockett, Laura K. Globig, Killian L. McLoughlin, Diana-Maria Mirea, Arthur Spirling, Diyi Yang, Steve Rathje, and Manoel Horta Ribeiro. 2026. “A Reporting Checklist for LLMs in Behavioural Science.” *Nature Human Behaviour*. <https://doi.org/10.1038/s41562-026-02492-7>.